BOOLEAN FUNCTIONS

Lecture 5

Boolean Functions

- A Boolean function is a quantity that depends on several other Boolean variables.

Example: $f(A,B,C) = A+B+C$

Note all the quantities A,B, and C are Boolean values, so can be either 1 or 0

Truth Table

- A truth table is a tabular representation of all the combinations of values for inputs and their corresponding outputs

- A breakdown of a Boolean function by listing all possible values the function can attain.

Truth Table (Continued...)

- How many Input possibilities a function can have?
  - 2 variables ?
  - 3 variables ?

n variables - 2^n possibilities

Truth Table (Continued...)

Q: Can you draw the truth table for $Y = A' + (B'.C)$ function?



$$Y = A' + (B'.C)$$

A	B	C	A'	B'	B'.C	Y
0	0	0	1	1	0	1
0	0	1	1	1	1	1
0	1	0	1	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	1	0	1	1	1
1	1	0	0	0	0	0
1	1	1	0	0	0	0

Minterms and Maxterms

Minterms and Maxterms

- Two ways of representation.
- Each input possibility of a Boolean function can be written either as a Minterms or as a Maxterms

How to obtain the Minterm?

- See the variable values,
  - If it is 1 → use the normal letter (A,B etc.)
  - If it is 0 → use the bar (A',B' etc.)
  - Combine them with AND.

- Examples
  - 000 → A'B'C' (Minterm representation of 000)
  - 011 → A'BC (Minterm representation of 011)

Q: Find the Minterms



A	B	C	Minterms
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

A	B	C	Minterms
0	0	0	(A'. B'. C')
0	0	1	(A'. B'. C)
0	1	0	(A'. B . C')
0	1	1	(A'. B . C)
1	0	0	(A . B'. C')
1	0	1	(A . B'. C)
1	1	0	(A . B . C')
1	1	1	(A . B . C)

How to obtain the Maxterm?

- Exactly the opposite procedure of obtaining Minterm
- See the variable values,
  - If it is 1 ⟶ Use the bar (A',B' etc.)
  - If it is 0 ⟶ Use the normal letter (A,B etc.)
  - Combine them using + operation

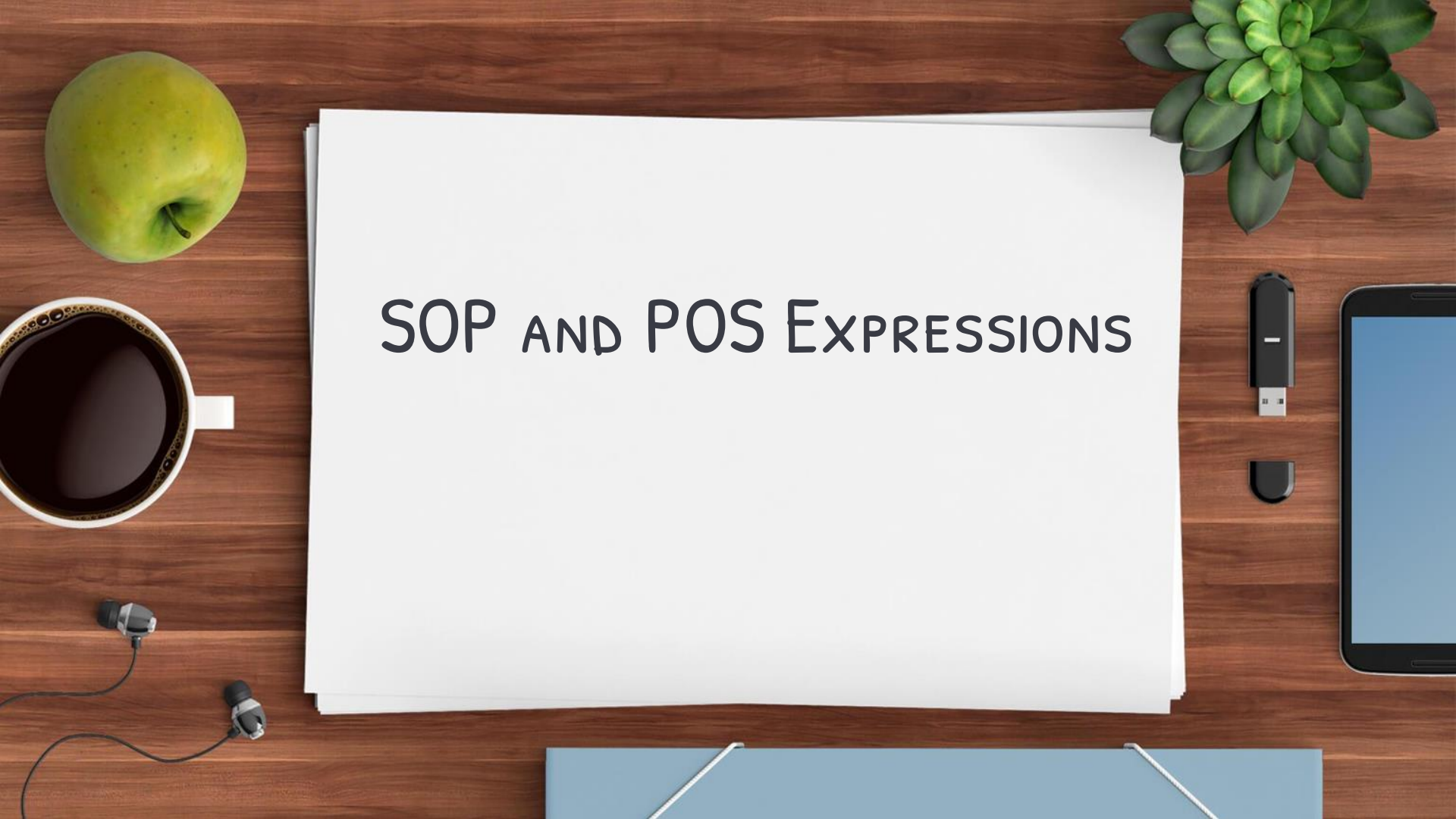
- Examples
  - 000 ⟶ A+B+C (Maxterm representation for 000)
  - 011 ⟶ A+B'+C' (Maxterm representation for 011)

Q: Find the Minterms



A	B	C	Minterms	Maxterms
0	0	0	(A'. B'. C')	
0	0	1	(A'. B'. C)	
0	1	0	(A'. B . C')	
0	1	1	(A'. B . C)	
1	0	0	(A . B'. C')	
1	0	1	(A . B'. C)	
1	1	0	(A . B . C')	
1	1	1	(A . B . C)	

A	B	C	Minterms	Maxterms
0	0	0	(A'. B'. C')	(A + B + C)
0	0	1	(A'. B'. C)	(A + B + C')
0	1	0	(A'. B . C')	(A + B' + C)
0	1	1	(A'. B . C)	(A + B' + C')
1	0	0	(A . B'. C')	(A' + B + C)
1	0	1	(A . B'. C)	(A' + B + C')
1	1	0	(A . B . C')	(A' + B' + C)
1	1	1	(A . B . C)	(A' + B' + C')


SOP and POS Expressions

- Now you know how to write each possibility of the input variables in two representations known as Minterms and Maxterms.

- Similarly, the Boolean function itself also can be written in two representations known as,
  - SOP → Sum Of Product (Use Minterms)
  - POS → Product Of Sum (Use Maxterms)

Note: Either you write a function in SOP form or POS form, it is the same function.

Sum of Product (SOP) Representation

1. First obtain the truth table
2. By looking at the truth table note down the places where the function becomes 1 (one)
3. Take only those possibilities and write them in Minterm representation
4. Combine these minterms using + operation
5. Now you have several products added together, so it is called Sum Of Products.

Q: Obtain the SOP expression for the following function ⟶



A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Example: Obtain the SOP expression for the following function →

$(A'.B'.C') + (A'.B.C') + (A'.B.C) + (A.B.C')$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Product of Sum (POS) form -

1. First obtain the truth table
2. By looking at the truth table note down the places where the function becomes 0(zero)
3. Take only those possibilities and write them in Maxterm representation.
4. Combine these minterms using . operation
5. Now you have several sums multiplied together, so it is called Product of Sums.

Q: Obtain the POS expression for the following function ⟶



A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q: Obtain the POS expression for the following function →

$(A+B+C').(A'+B+C).(A'+B+C').(A'+B'+C')$

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Thanks!

